Suggested Teaching Guidelines for

Deep Neural Networks

PGCP-AI February 2026

Duration: 40 hours Theory + 50 hours Lab + 30 Hours Self-Learning

Objective: Deep Neural Network

Prerequisites: Knowledge of fundamentals of programming and basic mathematics & statistical knowledge.

Evaluation method: Theory exam– 40% weightage

Lab exam – 40% weightage

Internal exam – 20% weightage

List of Books / Other training material

Courseware:

Deep Learning using Python, by S Lovelyn Rose, L Ashok Kumar, D Karthika Renuka, Wiley India, 1st Ed

Reference Book:

1. Deep Learning with Python by Francis Chollet
2. Deep Learning, Ian Goodfellow, Yoshua Bengio, Aaron Courville
3. Neural Networks and Learning Machines, Simon Haykin
4. Pattern Recognition and Machine Learning, Christopher M. Bishop
5. Hands-On Machine Learning with Scikit-Learn and TensorFlow
6. TensorFlow Deep Learning Cookbook
7. Reinforcement Learning with TensorFlow: A Beginner's Guide to Designing Self-learningSystems with TensorFlow and OpenAI Gym SayonDutta
8. Hands-On Reinforcement Learning with Python: Master Reinforcement and Deep Reinforcement Learning Using OpenAI Gym and TensorFlow Sudharsan Ravichandiran
9. Deep Reinforcement Learning Hands-On: Apply Modern RL Methods, with Deep Qnetworks, Value Iteration, Policy Gradients, TRPO, AlphaGo Zero and More MaximLapan

Session 1: (2T + 2L + 1SL)

Theory:

Introduction to Deep Neural Network

- Neural Network and its applications,Single layer neural Network
- Activation Functions: Sigmoid, Hyperbolic Tangent, ReLu ,Overview of Backpropagation of errors

Lab Assignment:

- Implement the different activation functions on the Dataset in the Jupyter notebook.

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Self-Learning:

- Explore why neural networks were needed beyond traditional ML
- Study Difference between:
    - Perceptron
    - Single-layer neural network
- Intuition behind activation functions:
    - Why linear functions fail
    - Why non-linearity matters

Session 2: (2T + 2L + 2SL)

Theory:

- Introduction to Tensorflow
- Introduction to Pytorch
- Comparison of Tensorflow and Pytorch

Lab Assignment:

- Explore Tensorflow ,Pytorch and their libraries.

Self-Learning:

- Study why deep learning frameworks exist
- Explore concept of Static graph vs dynamic graph
- Research vs production mindset:
    - Why PyTorch is preferred in research
    - Why TensorFlow dominates production

Session 3 & 4: (4T + 4L + 2SL)

Theory:

- Introduction Deep Learning and Neural Networks
- Practical Application of Neural Networks
- "Hello World" of Neural Network (Logistic Regression)
- Cost Function
- Activation
- Gradient Descent for Logistic Regression
- Back propagation

Lab Assignment:

- Implement the main steps of a Shallow Neural Network
    - Understand the dataset
    - Implement your first Forward and Backward propagation
    - Implement activation function, gradient descent
    - Build Neural Network Model
    - Test and optimize the model

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- Make Predictions

(to be done in Jupyter notebook. Need Jupyter-notebook and libraries installed)

Self-Learning:

- Justify why logistic regression is the “Hello World” of deep learning
- Explore Meaning of:
  - Cost function
  - Loss vs cost
- Intuition behind gradient descent:
  - Why slope matters
  - Why learning rate matters

Session 5: (2T + 4L + 2SL)

Theory:

- Sigmoid Model
- Sigmoid Loss function
- Introduction to Learning Algorithm
- Deriving the Gradient Descent Update Rule
- Sigmoid Evaluation

Lab Assignment:

- Plotting Sigmoid 2D
- Plotting Sigmoid 3D
- Contour Plot
- Plotting Loss
- Standardization
- Test/Train split

Self-Learning:

- Why sigmoid saturates
- Relationship between:
  - Prediction error
  - Loss curve shape
- Importance of standardization

Session 6: (2T + 4L+2SL)

Theory:

- Shallow Neural Networks
- Hidden Units and Hidden Layers
- Activation function- Tan-h and ReLu

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- Forward and Backward propagation with a hidden layer
- Deep Learning notations and Neural Network Representations

Lab Assignment:

- Implement a two-class neural network with a hidden layer
- Implement forward and backward propagation
- Compute the cross-entropy loss

Self-Learning:

- Explore why hidden layers are needed
- Study How hidden units learn representations
- Meaning of NN notation:
  - Weights
  - Bias
  - Activations

Session 7 & 8: (4T + 4L + 2SL)
Theory:

- Parameters vs Hyper Parameters
- Regularization
- L1-L2 normalization
- Frobenius norm

Lab Assignment:

- Implement L1-L2 Regularization
- Improve performance of the learning algorithms

Self-Learning:

- Study difference between:
  - Parameters
  - Hyperparameters
- Explore Overfitting intuition
- Explain Why L1 creates sparsity
- Explain Why L2 smooths weights

Session 9: (2T + 2L + 2SL)
Theory:

- Dropouts, Early Stopping
- Data Augmentation

Lab Assignment:

- Implement Dropouts, Early Stopping and Data Augmentation.

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Self-Learning:

- Explain why neural networks memorize
- Research Dropout as ensemble learning
- Study Early stopping as bias–variance tradeoff
- Explore why data augmentation helps small datasets

Session 10: (2T + 2L + 2SL)

Theory:

- Vanishing Gradient and Exploding Gradient problems
- Gradient Checking

Lab Assignment:

- Implement Gradient problems

Self-Learning:

- Explore why gradients vanish or explode and Study it layer by layer
- Study Effect of deep networks on gradients
- Understand why initialization matters

Session 11: (2T + 4L)

Theory:

- Batch Normalization and other methods for data normalization

Lab Assignment:

- Programming exercise for Normalization

Session 12: (2T + 2L)

Theory:

- Optimization Algorithms, ADAM
- Mini batch gradient descent

Lab Assignment:

- Implement ADAM in python and Tensor flow

Self-Learning: (2SL) (Note: Self-Learning including session 11 & 12)

- Research Internal covariate shift
- Study why batch normalization stabilizes learning
- Understand Difference between data normalization and batch normalization

Session 13: (2T + 2L + 2SL)

Theory:

- RMSProp, Momentum and other gradient descent algorithms

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Lab Assignment:

- Implement gradient descent algorithms

Self-Learning:

- Study why vanilla gradient descent is slow
- Explore and understand Intuition behind:
  - Momentum
  - RMSProp
  - Adam
- Research about Bias correction in Adam

Session 14 & 15: (4T + 4L + 2SL)
Theory:

- Tensor flow data structures and Library

Lab Assignment:

- Fashion MNIST and digits MNIST exercises using Tensor flow.

Self-Learning:

- Study and understand TensorFlow computation graph intuition
- Research and explore Difference between:
  - Model definition
  - Compilation
  - Training
- Understand why MNIST is still used

Session 16: (2T + 4L + 2SL)
Theory:

- Sequential Modelling, Introduction
- Basic building blocks, applications
- Introduction to RNNs

Lab Assignment:

- Implement RNN in Jupyter Notebook.

Self-Learning:

- Explain why feedforward networks fail on sequences
- Study Temporal dependency intuition
- Explore and understand use cases of Vanishing gradient in RNNs

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Session 17 & 18: (4T+4L+3SL)

Theory:

CNN - Convolutional Neural Networks

- Introduction, basic building blocks,
- Convolutional Concept
- Inception Network
- Transfer Learning
- Data Augmentation
- Padding, strides
- Pooling layers
- Fully connected
- Applications and Use Cases

Lab Assignment:

- Using tensor flow and python implement CNN for ASL (American Sign Language)

Self-Learning:

- Explore Limitations of fully connected networks for image data and need for CNNs
- Study Convolution operation, padding, strides, and pooling concepts
- Study and understand CNN architecture flow including fully connected layers and Inception networks
- Research about Transfer learning, data augmentation, and real-world vision applications

Session 19: (2T + 4L + 2SL)

Theory:

- GANs - introductions, basic building blocks
- Implementing GAN

Lab Assignment:

- Implement GAN using python and tensor flow.

Self-Learning:

- Explain Purpose of generative models and motivation for GANs
- Study Generator–discriminator architecture and adversarial learning concept
- Study GAN training challenges such as instability and mode collapse
- Research and understand Applications of GANs in data generation and augmentation

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Session 20: (2T + 2L + 2SL)

Theory:

Tuning Deep Learning Models

- Deciding Number of Layers
- Deciding Number of Neurons
- CNN tuning for better performance
- RNN/LSTM for Time Series Prediction

Trends in Deep Learning

- Echo State Networks / Reservoir Computing
- Auto Encoders
- Convolutional Auto Encoder
- Extreme Learning

Deep Learning Case Studies

- Whale Identification (5k Classes)
- Iris Detection
- Devnagari Digit Detection
- Flood prediction in Dam
- Heart Disease

Lab Assignment:

- Implement CNN using python and tensor flow
- Implement RNN/LSTM using python and tensor flow

Self-Learning:

Explore Reinforcement Learning: Markov Decision Processes, Dynamic Programming, Monte Carlo,Temporal Difference Learning, Approximation Methods